

Bharathi V

FRONTEND ENGINEER | REACTJS | JAVASCRIPT | REDUX TOOLKIT

Chennai, India | +91 8870947017 | bharathiviji1999@gmail.com
[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Frontend Engineer with **4+ years of experience** building and scaling React-based enterprise applications in complex, high-impact environments. Specialized in architecture-driven UI development, performance optimization, and state management, with hands-on experience modernizing legacy platforms into modular, scalable systems. Passionate about delivering high-performance, accessible, and maintainable user experiences that drive product velocity and measurable business outcomes.

PROFESSIONAL EXPERIENCE

Front End Developer - Tata Consultancy Services

Jul 2023 - Present

- Architected and developed scalable React-based enterprise modules for client **Tata Steel's** legacy application (Compass) and its modernized platform (CompassNXT) as part of a large-scale legacy-to-modern transformation initiative.
- Delivered major **bundle-size optimizations** by implementing **lazy loading**, **Webpack code splitting** and **asset restructuring**, reducing **initial entripoint size by ~59%** (5.90 MiB to 2.38 MiB) and improving **build time by ~72%** (165,704 ms to 44,983 ms), significantly enhancing load performance and developer productivity.
- Optimized rendering performance using **React.memo**, **useMemo** and **useCallback** to prevent unnecessary re-renders in complex UI workflows.
- Leveraged **Redux Toolkit** for centralized state management and **Context API** for scoped state handling to minimize prop drilling and enhance component reusability.
- Built modular, reusable **component systems** that reduced feature development turnaround time and improved long-term maintainability.
- Improved UI responsiveness by restructuring component hierarchies and refining dependency management in **React Hooks**.
- Contributed to **Backend-for-Frontend (Node.js + Express)** layer modifications, enabling seamless communication between UI and Java backend services.
- Deployed and monitored applications on **GCP Compute Engine VMs**, analyzing production logs through secure **SSH** access to diagnose and resolve issues efficiently.
- Integrated and optimized **RESTful APIs** within React applications, ensuring efficient state synchronization and data consistency.
- Supported **CI/CD automation via Jenkins**, ensuring stable builds and streamlined deployments.
- Ensured **accessibility (a11y) compliance** through **semantic HTML**, **ARIA roles**, keyboard navigation support and focus management best practices.
- Conducted API validation and debugging using **Postman** and **Swagger**, reducing integration issues during releases.
- Collaborated cross-functionally with backend, QA, DevOps and product stakeholders to ensure high-quality releases in **Agile environments**.
- Mentored team members on **UI architecture patterns**, **performance best practices** and scalable project structuring.

- Experienced front-end developer skilled in **React, JavaScript, HTML, CSS, Redux Toolkit, Tailwind CSS and Bootstrap**, with a strong focus on building scalable and responsive user interfaces.
- Possesses working knowledge of **PHP, Core Java and MySQL**, enabling effective collaboration with backend systems and APIs.

SKILLS SUMMARY

Technical Skills

- **Frontend:** React.js, JavaScript (ES6+), Redux Toolkit, Context API, React Hooks (useEffect, useMemo, useCallback, useRef), React.memo, TailwindCSS, SCSS, Material UI, Bootstrap, HTML5, CSS3
- **Performance & Optimization:** Code Splitting, Lazy Loading (React.lazy), Memoization, Bundle Optimization, Lighthouse Debugging, Accessibility (a11y), SEO, Semantic HTML, ARIA
- **API & Backend Integration:** REST APIs, Node.js (BFF - Express.js), JWT Authentication, Client-side Session Management, Postman, Swagger
- **DevOps & Cloud:** Jenkins (CI/CD), Git, GCP (Compute Engine, VM Instances), SSH, Production Log Analysis
- **Tools:** Webpack, Environment Variables (.env), Chrome DevTools

PERSONAL PROJECTS

- **Invoice-CRUD** - [View Project](#) | [Source Code](#)
 - **Tech Stack:** React.js, Material UI, Framer Motion, Bootstrap
 - Built a responsive invoice management application implementing full CRUD functionality with optimized state handling and smooth UI animations
- **Designo** - [View Project](#) | [Source Code](#)
 - **Tech Stack:** React.js, CSS3 (Custom Pure CSS Architecture), Framer Motion, Leaflet
 - Developed a UI-focused React application featuring handcrafted layouts using pure CSS, animation-driven interactions with Framer Motion and interactive maps powered by Leaflet.

EDUCATION

College of Engineering Guindy, Anna University
B.E. Agricultural & Irrigation Engineering

Chennai, India
Aug 2017 - Apr 2021